ETSN11, Final Exam 2025-10-30, Solutions

This document includes our proposed solutions. The solutions are meant as reference solutions to help you understand the expectations. Other solutions can also be correct.

If you find any errors or typos, please inform Maria Kihl (maria.kihl@eit.lth.se).

General advice: In **Part A,** questions 6-8, you need to correctly describe the specific terms and concepts to get full points. In **Part B**, you need to motivate your chosen formulas and calculations to get full points. In **Part C**, you need to give a suitable solution to the specific scenario. To get full points, this solution must both address the requirements and challenges in the scenario **and** include specific protocols and technologies.

Part A: Technologies and architectures (30 points)

In the **multiple-choice questions**, you get points for each correct answer, but **minus 2 points** for each incorrect answer. You cannot get a lower score than 0 in the multiple-choice questions. In the **text questions**, you need to correctly describe the specific terms and concepts to get full points.

1. Which of the following technologies and concepts are suitable for **resource-constrained networking**? (5p)

a. BLE Mesh
b. ZigBee
c. WiFi 6 Mesh
d. NB-IoT
e. Starlink
f. 5G URLLC
g. LTE-M
h. Argos
i. IEEE 802.11
j. LoRa
k. AODV
l. DTN
m. 5G eMBB
n. 5G mMTC
o. OFDMA

2. Which of the following statements are true for **ad-hoc networks**? (5p)

a. They require contention-free MAC protocols.
b. They require contention-based MAC protocols.
c. They can use centralized network architectures.
d. They can use decentralized network architecture.
e. They can be used for time-critical applications.
f. They can be used for non-critical applications.
g. They can have a DTN architecture.
h. They require low communication error rates.

i. The nodes in an ad-hoc network always require low energy consumption.
j. There is always only one gateway in the network.
k. There can be multiple gateways in the network.
l. They always use radio-based communication links.
m. LoRa is an ad-hoc network.
n. ZigBee is an ad-hoc network.

3. Which of the following statements are true? (5p)

a. Communication links in outer space can use acoustic communication.
b. Communication links in outer space can use optical communication.
c. Underwater communication links can use radio communication.
d. Space relays are used as base stations on planets.
e. Outer space networks often have solar-powered systems.
f. Underwater networks often have solar-powered systems.
g. Space relays are used to re-direct communication links.
h. Space relays are used as base stations on planets.
i. A DTN node in space is always placed on a planet.
j. DTNs in outer space always use scheduled routing.
k. The Bundle protocol is a DTN protocol.
l. JANUS is a DTN protocol.
m. A space link provides one-way communication between two nodes.
n. A space link provides two-way communication between two nodes.

4. Describe the purpose, architecture and main functions of **Bluetooth.** (5p)

Solution: Bluetooth is a **decentralized (ad-hoc) network technology** aimed at Machine-to-Machine (M2M) communication. Bluetooth operates on the **unlicensed frequency band** on 2.4GHz. To avoid disturbances, Bluetooth communication uses frequency hopping. Bluetooth allows devices to connect peer-to-peer in a so called **piconet**. To set up a piconet, Bluetooth has a **pairing process**, which often involves user interaction. The piconet is controlled by the **primary** device, and other nodes in the piconet are **secondary** devices that follow the instructions from the primary device. Two nodes in a piconet can change roles at any time.

5. Describe the main functions for **DTNs** and describe an example of a network system based on a DTN architecture**.** (5p)

Solution: Delay- and Disruption Tolerant Networks (DTNs) are used to provide reliable data transfer in **network systems with intermittent connectivity**. For example, in environments where the connectivity is unpredictable or has a high error rate. A DTN uses the **store-and-forward** principle, where a node stores data (for example sensor data) until it reaches another node to pass on the data. A DTN **must include moving nodes** to be able to forward data. When the node has reached the sink of the network, which is connected to a gateway, the data can be transmitted through the backhaul to the internet.

For full points, you then need to give a concrete and technically correct example of a network system that is based on a DTN architecture. It can be an example from the course, or a system you have designed yourself.

Example: One example of a DTN system is a space network, which can have especially high latency and unstable links. In this system, DTN routers can be placed on satellites (moving nodes) and on ground stations on Earth or other planets. When a satellite is within range of a ground station, data can be transmitted. The DTN routers can store the data and forward the data when a link is available.

Example: One example of a DTN network is a sensor network, where sensors are placed on (mobile) animals and where there are stationary gateways that are used as sinks. A sensor collects and stores data about the animal. When the animal comes close to a gateway, the sensor can forward its stored data to the gateway.

6. Describe the main purpose and functions of **Backhaul links** and give at least three examples of such links. (5p)

Solution: The main purpose of a backhaul is to **connect an access network to the rest of the Internet**. It functions as the data link between the access network's gateway and a core network. Three examples of typical backhaul links seen in the course are wired links typically **Ethernet** (copper or fiber), **microwave links** (or a point-to-point radio link), and **Starlink**.

We have also used backhaul links based on cellular networks, but if you use this as an example, you need to describe the context, since a cellular network is not a backhaul link. There is a backhaul link from a cellular network base station, but this is typically wired. Also, you must describe specific technologies, not just general communication technologies.

Part B: Performance analysis problems (30 points)

7. You design a wireless access network including a base station and several user devices, see the illustration below. You decide to use an MAC protocol based on **slotted ALOHA**. In slotted ALOHA, user devices share a common radio channel, but a user device can only send data frames at specific time instants, called slots. If more than one user device sends a data frame in the same slot, the data is corrupted and lost. When the base station correctly receives a data frame, it sends an ACK to the sender on another radio channel.

There are 4 user devices, A-D.

- For each slot, user devices A and B send data frames with probability 0.2, and user devices C and D send data frames with probability 0.3.
- Assume that ACKs always are received before the next time slot has started. The probability that an ACK from the base station is received correctly by the user device is 0.8.
- All user devices and data transmissions are independent of each other.

a. What is the probability that a slot is empty, i.e. none of the user devices send a data frame in this slot? (2p)

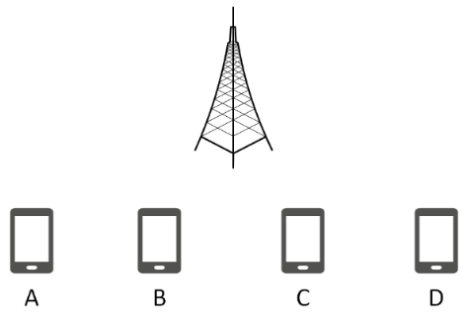
b. What is the probability that there is a data collision in a slot? (2p)

c. Assume that user device B sends a data frame in a slot. What is the probability that this data frame will be received correctly by the base station? (3p)

d. What is the efficiency of the MAC protocol, including the ACKs? (4p)

Assume that the application on user device B includes a **Stop-and-Wait ARQ protocol**. If the user device does not receive an ACK when it has sent a data frame, the user device retransmits the data frame in the next slots until it receives an ACK. We can assume that the other user devices send data frames with the same probabilities as above.

e. Assume that user device B wants to send a data frame. What is the average number of required slots for a successful transmission of this data frame? (4p)



Solution:

This is a rather straightforward exercise if you know the basic probability laws and understand what you should calculate. There are many ways to make a shorter solution than the solution suggestion shown below.

What do we know?

For each slot: P(A sends a frame) = P(B sends a frame) = 0.2 and P(C sends a frame) = P(D sends a frame) = 0.3. For each slot, there are the following outcomes (No = Device X does not send a frame; Yes = Device X sends a frame):

Nr	A	B	C	D	P(outcome)
1	No	No	No	No	$0.8 \cdot 0.8 \cdot 0.7 \cdot 0.7 = 0.314$
2	Yes	No	No	No	$0.2 \cdot 0.8 \cdot 0.7 \cdot 0.7 = 0.0784$
3	No	Yes	No	No	$0.8 \cdot 0.2 \cdot 0.7 \cdot 0.7 = 0.0784$
4	No	No	Yes	No	$0.8 \cdot 0.8 \cdot 0.3 \cdot 0.7 = 0.134$
5	No	No	No	Yes	$0.8 \cdot 0.8 \cdot 0.7 \cdot 0.3 = 0.134$
6	Yes	Yes	No	No	$0.2 \cdot 0.2 \cdot 0.7 \cdot 0.7 = 0.0196$
7	Yes	No	Yes	No	$0.2 \cdot 0.8 \cdot 0.3 \cdot 0.7 = 0.0336$
8	Yes	No	No	Yes	$0.2 \cdot 0.8 \cdot 0.7 \cdot 0.3 = 0.0336$
9	No	Yes	Yes	No	$0.8 \cdot 0.2 \cdot 0.3 \cdot 0.7 = 0.0336$
10	No	Yes	No	Yes	$0.8 \cdot 0.2 \cdot 0.7 \cdot 0.3 = 0.0336$
11	No	No	Yes	Yes	$0.8 \cdot 0.8 \cdot 0.3 \cdot 0.3 = 0.0576$
12	Yes	Yes	Yes	No	$0.2 \cdot 0.2 \cdot 0.3 \cdot 0.7 = 0.0084$
13	Yes	Yes	No	Yes	$0.2 \cdot 0.2 \cdot 0.3 \cdot 0.7 = 0.0084$
14	Yes	No	Yes	Yes	$0.2 \cdot 0.8 \cdot 0.3 \cdot 0.3 = 0.0144$
15	No	Yes	Yes	Yes	$0.8 \cdot 0.2 \cdot 0.3 \cdot 0.3 = 0.0144$
16	Yes	Yes	Yes	Yes	$0.2 \cdot 0.2 \cdot 0.3 \cdot 0.3 = 0.0036$

All outcomes are independent of each other and mutually exclusive.

Also, we know that when a frame is received correctly by the base station, the base station sends an ACK.

P(an ACK is received correctly by a device) = 0.8

a. What is the probability that a slot is empty, i.e. none of the user devices send a data frame in this slot?

Solution:

P(a slot is empty) = P(no device sends a frame) = P(outcome 1) = 0.314

b. What is the probability that there is a data collision in a slot?

Solution:

P(there is a data collision in a frame) = P(more than one device sends a frame) = P(outcome 6-16) = 1 – P(outcome 1-5) = 0.261

c. Assume that user device B sends a data frame in a slot. What is the probability that this data frame will be received correctly by the base station?

Solution:

In this case, we know that device B has sent a frame, that is, we are only interested in outcomes 3, 6, 9, 10, 12, 13, 15, and 16.

Nr	A	B	C	D	P(outcome | B has sent a frame)
3	No	Yes	No	No	$0.8 \cdot 0.7 \cdot 0.7 = 0.392$
6	Yes	Yes	No	No	$0.2 \cdot 0.7 \cdot 0.7 = 0.098$
9	No	Yes	Yes	No	$0.8 \cdot 0.3 \cdot 0.7 = 0.168$
10	No	Yes	No	Yes	$0.8 \cdot 0.7 \cdot 0.3 = 0.168$
12	Yes	Yes	Yes	No	$0.2 \cdot 0.3 \cdot 0.7 = 0.042$
13	Yes	Yes	No	Yes	$0.2 \cdot 0.3 \cdot 0.7 = 0.042$
15	No	Yes	Yes	Yes	$0.8 \cdot 0.3 \cdot 0.3 = 0.072$
16	Yes	Yes	Yes	Yes	$0.2 \cdot 0.3 \cdot 0.3 = 0.018$

P(the data frame is received correctly by the base station) = P(outcome 3 | B has sent a frame) = 0.392

d. What is the efficiency of the MAC protocol, including the ACKs?

Solution:

The efficiency of the MAC protocol can be seen as the ratio (probability) of "useful" slots, that is slots with a successful data transfer (i.e, an empty slot is not useful). In this case, we should also include the ACKs, which means that a slot is useful if a data frame sent in the slot is received correctly by the base station AND the corresponding ACK is received correctly by the sender.

P(a slot is useful) = P(outcomes 2-5) · P(ACK is received correctly) = 0.34.

e. Assume that user device B wants to send a data frame with Stop-and-Wait ARQ (as described above). What is the average number of required slots for a successful transmission of this data frame?

Solution:

A data frame sent by B is correctly received by the base station if none of the other devices send any frames.

P(base station correctly receives a data frame from B) = P(outcome 3 | B sends a frame) = 0.392

P(B sends a frame in a slot and receives an ACK) = P(successful data transmission) = 0.392 · 0.8 = 0.314

Let X = number of slots until a successful transmission.

$P(X = k) = (1 - 0.314)^{k-1} \cdot 0.314$ (k-1 unsuccessful transmissions and 1 successful transmission), that is X follows a geometric distribution.

$E[X] = \frac{1}{0.314} = 3.2$ (from formula sheet)

Since E[X] is an average value, it does not have to be a whole value (integer). But it is ok to answer 4 slots as well.

Note: If you have not included the ACK in your calculation, you have received 0 points on this question.

8. You are designing a new **application protocol**. The application protocol receives data packets with information from the transport protocol. The information should be processed by the application and then shown in the graphical user interface (which means that the data packet leaves the application protocol).
The arrivals of data packets can be modelled as a Poisson process with mean 2 packets per second. The average processing time for one data packet is 0.25 seconds. All packets and processing times are independent of each other. (15p)

a. What is the probability that exactly 2 new data packets will arrive at the application protocol during an interval of one second? (2p)

   Assume that you first model the application protocol process as one worker thread with an unlimited job queue. If the worker thread is busy when a data packet arrives, the data packet is placed in the queue. The data packets in the queue are processed first-come-first-served by the worker thread.

b. Assume that the processing times are modelled with an exponential distribution. What is the average response time for a data packet? (3p)

c. Assume that the processing times are modelled with a deterministic distribution. What is the average response time for a data packet? (3p)

   Assume instead that you model the application protocol as 4 parallel worker threads that independently can process data packets. If there is an idle worker thread when a data packet arrives then the packet is processed (the work is randomly shared between worker threads). If all worker threads are busy when a data packet arrives, the data packet is blocked and leaves the system. The processing time for a data packet that enters a worker thread can be modelled as an exponential distribution with mean 0.25 seconds.

d. What is the probability that a data packet is blocked? (3p)

e. What is the average throughput of processed data packets? (4p)

Solution:

This is a straightforward queueing system problem, but in another context.

a. What is the probability that exactly 2 new data packets will arrive at the application protocol during an interval of one second?

   Solution:

   New data packets arrive according to a Poisson process with mean 2 packets per second.

P(2 packets arrive during an interval of one second) = $e^{-2} \cdot \frac{2^2}{2!} = 0.27$

(Formula for Poisson distributed variables found in formula sheet)

In exercises b and c, the system can be modelled as an M/G/1 system with unlimited queue.

λ = 2 requests per second and E[X] = 0.25 seconds (μ=4).

This means that $\rho = \frac{\lambda}{\mu} = 0.5$.

b. Assume that the processing times are modelled with an exponential distribution. What is the average response time for a data packet?

Solution:

Exponentially distributed processing times means that the system can be modelled as an M/M/1-system.

Average response time for a data packet = $T = \frac{1}{\mu(1-\rho)} = 0.5$ seconds (Found in the formula sheet).

c. Assume that the processing times are modelled with a deterministic distribution. What is the average response time for a data packet?

Solution:

A deterministic processing time means that Var(X)=0.

For M/G/1-systems, the Pollaczek-Kinchine formula can be used (found in the formula sheet):

$\bar{N} = \rho + \frac{\rho^2+\lambda^2 \cdot Var(X)}{2 \cdot (1-\rho)}$ where $\bar{N}$ is the average number of data packets in the system.

Then use Little's theorem to find the average response time for a data packet.

$T = \frac{\bar{N}}{\lambda} = \frac{1}{\mu} + \frac{\rho^2+\lambda^2 \cdot Var(X)}{2 \cdot \lambda \cdot (1-\rho)} = 0.25 + \frac{0.25}{2} = 0.37$ seconds.

In exercises d and e, the system can be modelled as an M/M/4-system without queue, that is an Erlang system.

λ = 2 requests per second and E[X] = 0.25 seconds (μ=4).

This means that the offered traffic $A = \frac{\lambda}{\mu} = 0.5$ (from the formula sheet).

d. What is the probability that a data packet is blocked?

Solution:

For Erlang systems, the Erlang loss formula can be used. The Erlang loss formula determines the probability that a job (data packet) in the system is blocked because the system is full when the data packet arrives at the system.

In this case:

$P_B = \frac{A^4/_{4!}}{\sum_{j=0}^{4} \frac{A^j}{j!}} = \frac{0.002604}{1.648} = 0.0016$ (Ok, maybe not the best choice of values in this exercise, but the principle is the important part here.)

e. What is the average throughput of processed data packets?

Solution:

The average throughput of processed data packets, also called λ_{eff}, is calculated as $\lambda_{eff} = \lambda \cdot (1 - P_b) = 1.997$ data packets per second, so basically all data packets are processed.

Part C: System design scenarios (30 points)

For each of the scenarios given below, there are some general recommendations for the system design in this scenario, and some solution examples, which were given full points on the exam. The solutions have been only minor edited, fixing typos and small errors that were not given point reductions.

Grading: You get full points if your system solution is realistic, includes all important parts, and your arguments are correct and coherent. You get 8-14 points if you have provided a coherent system design with correct network technologies, but your solution lacks some details (the reductions depend on what you lack in your solution). You get 0-7 points if your chosen network technologies are not suitable, your solution lacks large parts of the important challenges, or your solution is general and thereby not including technical details for this specific scenario. Further, you need to suggest **one solution**, not provide different alternatives. If your answer provides different alternatives (or if this is not clear), you will get point reductions. If your solution is not coherent or not technically correct in all parts, you get point deductions.

9. Foodora has started a trial with food-delivery robots together with Stockholm municipality, see figure below. About 15 Foodora robots are driving around the area of Södermalm, see map below, delivering food to customers. Currently, the robots are autonomous, using sensors and cameras to avoid crashing into people and other things in their way. The robots have a range of about 3 kilometers. When they have delivered food, they go back to the main central placed in the middle of Södermalm for re-charging and receiving instructions for new deliveries.

Currently, robots are fully autonomous and self-driving, meaning that robot AI is computed in the robot itself. However, Foodora would like to develop a solution for cloud robotics, meaning that the robots should be connected to a network and that robot AI computing is performed in the cloud instead of in the robot itself.

Your task is to help Foodora with this cloud robotics solution. The robots have sensors and cameras, but all computing should be placed in a cloud-based back end. You must suggest a complete network design solution that is suitable for the Foodora food-delivery robots. The robots will continue to operate in Södermalm. You have a decent budget for your solution and all available required technologies.





Scenario: Foodora are trying out robot deliveries on Södermalm, read more here (in Swedish). These robots are autonomous, so they do not need a network connection for AI computations.

Recommendations: This scenario represents a typical time-critical application. The scenario is situated in the middle of Stockholm, so there are good opportunities to use already existing infrastructure.

For full points, you need to describe a suitable wireless network technology for the robots, a solution for the localization of robots (required for a safe solution), a realistic solution for the cloud back-end, and a solution for power supply. Also, you need to describe how the different parts form a complete end-to-end networked application. You do not get extra points for describing things that are not directly relevant for the system design.

Using the existing 5G public networks for wireless connection is the most straightforward solution. An integrated private 5G network using network slicing can be one option. Solutions proposing that Foodora should deploy an independent private 5G network or that Foodora will buy a 5G frequency band and become its own operator are **not** suitable, since they are very expensive and not needed in this scenario. Starlink-based solutions are accepted if they are technically correct. Solutions including network technologies aimed at non-critical applications are not suitable.

To get full points, you need to describe a suitable and technically correct solution for the cloud back-end. Your solution must show that you understand the time-critical requirement. The easiest solution is that Foodora deploys a private cloud in the main central (or somewhere in Stockholm area). Solutions that use public cloud services are accepted as well. If you suggest edge or fog computing, you need to describe how these solutions are deployed. If you do not describe how Foodora should deploy the cloud back-end, you get point deductions.

For localization, GPS is the easiest solution, since it is an outdoor scenario. A 5G connection can include a position, but then you need to describe this to get points.

For power supply, the robots have batteries that can be re-charged at the main central.

Solution example 1:

This is a time-critical scenario which requires low latencies and high capacity since its a scenario about autonomous driving. A suitable network technology for this scenario is 5G. This is because Södermalm is in Stockholm which is a big city with good 5G coverage. With 5G, Foodora can get their own private network by using network slicing. This would mean that they pay a provider to reserve capacity in their public 5G network, this would decrease interference and would improve the overall communication. 5G also supports URLLC which means ultra reliable low latency communication which is essential for autonomous driving.

For this solution to work, each Foodora robot would be equipped with their own eSIM so that different robots can be identified. Their location can be tracked with GPS so that both

the customers as well as the cloud and Foodora employees know the whereabouts of the robots.

The server for cloud computing can be placed in the main central where the robots charge, the server would be mains powered here. They could also use edge computing and spread-out devices to bring cloud computing closer to robots, in that way reducing latency. Edge computing could also be a beneficial solution because it would reduce the amount of work the main server would have. However, this means that Foodora would need to find safe places around Södermalm where the edge devices can be mains powered. Since this is a city it's very possible but it is a little more expensive and requires a little more work since they must find places to put the edge devices.

Since the robots charge in the main station, there is no need for energy harvesting. They could have solar panels but since Sweden does not have a lot of sun, they only go on short trips, and can be charged frequently, it seems expensive and unnecessary.

Solution example 2:

This scenario requires following:

- Self-driving food delivery robots
- All computing in a cloud
- Reliable connection with low latency between cloud and robots
- Some form of positioning technique, so that the robots can send their position to the cloud.

This scenario is a typical time-critical application where reliable data delivery and low latency is crucial. Robots are self-driving vehicles, and therefore it's important that they remain a stable connection with fast response from the computing cloud so they don't do something unexpected, such as driving out in the streets, crashes etc. From the instructions, we know that this project is located in Södermalm, Stockholm, and can therefore assume that there is good cellular coverage. Before, the robots were fully autonomous and self-driving, meaning that robot AI was computed in the robot itself, but now Foodora wants to move these calculations to a cloud. The robots are equipped with sensors and cameras to avoid crashing into things, and there are about 15 robots driving around.

A good approach for this scenario is to take advantage of the existing cellular coverage and use 5G connections. 5G is specifically good in this scenario, since it enables URLLC, ultra reliable low latency communication. Since this project has the objective that Foodora should have reliable food delivery, I suggest that the company invest in an integrated private 5G network. This means that they buy capacity from the existing cellular infrastructure, to make sure that all robots can always have enough stable connectivity. The frequency bands for the cellular network would still be managed by the operator but would not affect the private network. The backhaul is provided by the mobile operator, and we can assume it´s a robust, reliable and low latency backhaul that sends data to the internet so the cloud can access it.

Furthermore, the cloud should be a private cloud, specifically aimed at this project and company. It is preferable that the computations are not located too far away from the target area, and therefore we could "move the cloud closer" to the users with edge computing. The

computations could be made in servers placed in the area of Södermalm, which would lower the latency of data transmissions and ensure a more reliable connection. The servers are mains powered, since there is plenty of electricity in the inner city of Stockholm, and have wired backhaul solution to internet via fiber.

The robots are battery powered devices and have a main central they go back to when they must recharge and get instructions for new deliveries. When robots are on a delivery mission, they must continuously send data to the cloud, so that computations can be made about driving path etc. The data being sent will mainly consist of their position, so that the cloud can compute where the robots should drive. GPS is a technology that could be suitable for this scenario, since it's used in many outdoor navigation scenarios. The robot can send its position and destination address, and the cloud could compute the best path for it. On the way, the robot will use its sensors and cameras to avoid collisions and detect obstacles on the way. If it's blocked for any reason, it could send an alarm, with new position info to the cloud so it makes new calculations for the route. Also, the cloud should get information about the battery power of a robot, so it ensures the robot will have enough battery power to both deliver and go back to recharge.

This solution is based on a centralized technology using an integrated private 5G network for reliable connectivity, low latency and support for mobility. The scenario is time-critical, requires fast responses with reliable connection and support for mobility. To fulfill those requirements, 5G is the best option. The computations are moved to a private cloud owned by Foodora and will get data from the vehicles continuously. To ensure that the computations are made close to the target area and reduce latency, it will take advantage of edge computing.

A public cloud could be used for this project, but I feel like there is easier for the company to expand the solution if they have a private one. It is more expensive, but the decent budget will cover it perfectly. Also, there is no need to set up a private stand-alone 5G network, since the target area is very big and the costs will increase extremely. Using WiFi 6/7 could be a solution, but since the robots have a long range, it would require a really big mesh network to cover the entire area. Also, WiFi does not support mobility in the same way as cellular coverage does, so if a robot drives out or range of communication to a WiFi node, it will stop working. Backhaul in this solution is provided by the mobile operator, probably in a wired form, and I assume my edge computing servers are also connected to Internet via wired backhaul, such as fiber. The main costs for this project are buying capacity for the integrated network, and the private cloud. Foodora could use a public cloud to lower the costs, but in the long term, I think it would benefit them to not do it.

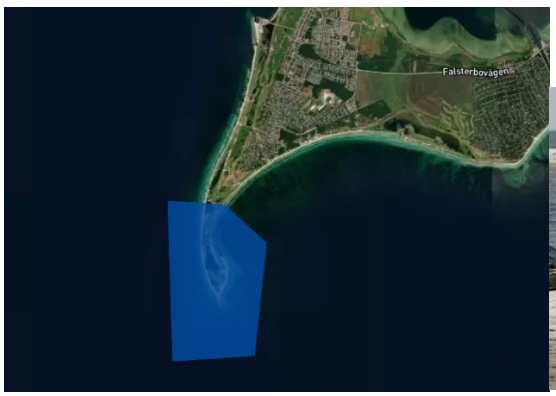
10. Måkläppen is a nature reserve that protrudes from the Falsterbo peninsula south of Malmö (see figures below). It was created by sandbanks built on top of a moraine core. The total size of Måkläppen (land + water) is about 8 km^2, where 0.3 km^2 is based on land and the rest is water.

    Måkläppen is mainly famous for the seals that live and breed there. Therefore, Måkläppen is only open for visitors during Nov 1 - Jan 31 to protect the seals. Unfortunately, lots of people do not respect these rules, and they try to visit the nature reserve at other times of the year as well, thereby disturbing the seals and their pups.

    Måkläppen Foundation, based in Falsterbo, manages the nature reserve. Your task is to help Måkläppen Foundation with a complete end-to-end network system that has two main objectives:

    1. Monitor the seal colony and learn more about the lifes of the seals. For example, Måkläppen Foundation wants to learn more on how deep the seals dive when they are looking for food and the seals' average swimming distance per day. You can assume that the seals stay within the Måkläppen reserve.
    2. Act as an alarm system to detect when people visit the reserve when it is closed for visitors. You can assume that people come on foot from Falsterbo or by boat.

    You have a relatively limited budget, since Måkläppen Foundation is a non-profit organization. You have access to different types of sensors that are animal friendly and that can measure depth and other relevant metrics. Also, the sensors include functions that can detect if they are above or underwater, and they can use GPS. Further, you have access to web cameras and other standard monitoring equipment. Finally, all sensors and equipment can use various wireless network technologies.





    Scenario: Måkläppen and its seals exist. There is a non-profit organization for Måkläppen. The rest of the scenario is fictional, but the general scenario of animal monitoring and trespassing detection systems is very realistic. Also, I removed one fact to make the scenario slightly easier: The land parts of Måkläppen constantly changes as sand builds up and is taken away by wind, ocean currents, storm surges and high tides.

Recommendations: This is a rather complex scenario, with two application objectives and a cost constraint. The whole scenario corresponds to a non-critical application, since neither the animal monitoring system nor the trespassing detection system require latencies in the range of milliseconds.

For full points, you need to provide a technically correct, coherent **and** cost-effective solution for the complete end-to-end networked system, including wireless network connectivity for monitoring and alarm system, backhaul, back-end, and describe a power supply solution for all included devices and network equipment.

Your solution must meet the requirements stated in the scenario. This means that the system must enable the Måkläppen foundation to learn more about the seals' lives, including the seals' underwater behavior, and send an alarm when people come by foot or by boat.

The cost-effectiveness of your solution will be part of the grading. You have learnt about many wireless network technologies during the course, some more cost-effective than others. For full points, you need to include some type of arguments for the cost of the system, showing that you have thought about this when you designed the solution.

For the alarm system, your choice of wireless network technology depends on how you design the system. There is 4G/5G in Falsterbo, and you can assume that there is good cellular connectivity in the areas close to Falsterbo. For the areas further out in the ocean, you can argue for cellular connectivity. It is not realistic to only have cameras on land that should cover the whole Måkläppen land and sea areas.

For seal monitoring, you need to propose a wireless sensor network that can handle that the seals are spending much time underwater. You need to describe how the sensors should be deployed. Argos is a very expensive technology, and Argos is not needed here since we are close to civilization. Private 5G networks are very expensive and not suitable. A private LoRa network, NB-IoT via a 4G/5G operator, or BLE mesh can be used. Note that a LoRa network requires a network server that must be placed somewhere. ZigBee is mostly aimed at indoor scenarios, since the range is limited. Also, if you propose ZigBee or BLE mesh, you need to describe the network architecture with different types of nodes. You can include a DTN architecture, where sensors store data underwater and send the data when they are above water, but then you need to describe how this is done.

Also, you need to propose a back-end server solution for the data. The scenario states that Måkläppen Foundation is located Falsterbo, so you can argue for that they have an office there.

For power supply, the sand banks of Måkläppen do not have an electricity grid, but the parts closer to Falsterbo can be assumed to have access to electricity.

All student solutions lacked something, but here are two example solutions that are almost complete. Both examples use a LoRa network, but NB-IoT would work fine as well.

Solution example 1 (Lacking some details about the back-end):

Because this is a monitoring, sensor scenario, we can classify this as a non-critical, resource-constrained application. Even though you of course want to know if there are trespassers as soon as possible, there is not a requirement of latencies of milli- or microseconds.

First of all, we have to locate where we want our sensors.

To fulfill the first objective, to monitor the seals, I propose that the seals are equipped with small, animal-friendly sensors. Because these sensors include functions that can detect if they are above or underwater, I suggest these nodes implement Store-and-Forward Message Switching functionality: making them collect data whenever they are underwater, about metrics such as which depth the seals are travelling. Whenever the sensor detects it is above water, it will try to transmit. If the sensors are also able to track the movement of the seals this data is of course also included, otherwise, the location when the seals are above water can be "snapshotted" with GPS and included. This would not of course bring the whole picture of the seals movements but give them a simplified picture at least.

To fulfill the second objective, to monitor the reserve of trespassers, I would strategically place sensors with camera functionality around the area, at the edges, and along the shore (to be able to detect both people on foot and by boat). These web cameras would be in sleep mode unless movement is detected, then a picture is taken and sent to the base stations. This data is then forwarded via a gateway onto the backhaul link to the network server, warning and making the picture in some way available for the Måkläppen Foundation, for example via a mobile phone application, which then can examine the photo to make sure it is actually a trespasser.

With these two types of sensors in mind, and the constraints of the project having a limited budget, my proposed solution is to build a LoRa-network. There might be a public LoRa network architecture available from the town of Falsterbo, otherwise, a private one can be set up. The base stations can then be strategically placed, for example at the edge between civilization and the nature reserve such that the base stations can be mains powered. Because of the long range LoRaWan provides, the distance from the seals' positions in the sea into the shore should not be a problem. If necessary, there can be LoRa base stations in the nature reserve, and these could be powered by some kind of energy harvesting, such as solar power, and a battery to store excessive power.

The sensors also need some type of power supply. Mainly, they will be battery powered. Since LoRa has low energy consumption, the battery life should be at least a few years. The sensors of the second objective is clearly class A LoRa devices, who will remain in sleep mode until movement is detected. The battery life could though be prolonged together with some type of energy harvesting; for the sensors monitoring the nature reserve, this could be done with solar power harvesting. For the seals, it could become a bit more complicated to have an energy harvesting solution while still ensuring they are animal friendly. However, one solution could be to use some kind of thermal energy harvesting from the water if possible.

The base stations also need some type of backhaul link. As earlier mentioned, the nature reserve is close to the town of Falsterbo, potentially allowing for the base stations to make use of already existing infrastructure, such as wired solutions. Otherwise, a microwave point-to-point backhaul link, connecting the nature reserve with the town, could be a potential solution.

Solution example 2 (Lacks details about the LoRa network server):

So, there are a few challenges here. First of all, we have to have monitor equipment that works underwater and a solution for how to send the data from the seals within a limited budget. As well as installing a alarm system, also within the range of a limited budget. This is not a critical scenario considering that we don't need to constantly monitor the seals but rather get updates every now and then to collect data and statistics. For the alarm system, this might seem like a time critical but in this case, it still isn't since we don't need updates at all the times - only when something has been detected.

For the monitoring of the seals, my idea is to put LoRa sensors on the seals to measure the depth, swimming distance and so on. LoRa is a non-critical network technology often used to monitor animals in large areas. LoRa sensors can send small packages of data but have low power consumption. Since we have sensors that can detect whether the seal is underwater or not (assuming that seals don't stay underwater for too long) my solution is that the sensors (which measure different factors) store the data and then the LoRa module sends it once the seal reaches the surface. LoRa uses radio waves to send data to their gateway which means it works best over the surface. The sensors will of course all be in one case and somehow put on the seal either like a necklace or on the fin (whatever impacts the animal the least). For power consumption, I will use batteries. LoRa has batteries that can last for up to 10 years which is perfect for this scenario where we don't want to disturb the animals and change the batteries often. To minimize the energy consumption, I want to use class A devices so that they sleep most of the time and only wake up once they are triggered (we reach the surface). This system design could of course be impacted in regard to how often a seal reaches the surface, but I will assume they don't do it very often so that the LoRa sensor can sleep most of the time.

Another solution if we had an unlimited budget would be to use Argos and have Argos "antennas" on the animals and then let the seals send data to a satellite, wherever they are. But this is not a good solution because of the limited budget since Argos can be expensive.

Once the animal reaches the surface, we want to send the data to a LoRa gateway. A LoRa gateway can reach up to 15 km^2 which covers this area and therefore we only need one gateway (which is good for the cost). Hopefully the gateway can be connected to the electricity grid and be mains powered, but a hybrid solution with solar cells is probably good (although we are in Sweden, so it has more cons from the environmental perspective). The backhaul for the gateway will be cellular network, assuming it exists. The gateway then forwards the data to a local server or the cloud so that the foundation then can access it and analyze the data.

For the alarm system I want to use the same LoRa gateway to minimize costs and therefore use LoRa sensors for this system as well. So, I will have buoys on the water alongside the nature reservation area in the water and on the ground have a stationary sensor where the people might walk into the nature reserve. Together with these sensors I will have movement detection and a camera which sends low quality images. Although LoRa can only send very small amount of data, since the pictures are of low quality I think it will still be able to send the pictures. Once a movement is detected the sensor will be triggered and a picture will be taken and sent to the gateway which then sends the picture further to the cloud or server. This way the foundation can see the image and scan it for either boats, people or if it was just an animal. Since the picture is sent when triggered, the foundation will have time to see the boat/person and ask them to leave. If possible, within the budget, It would be good with a processor in the cloud that can see whether the picture seems to be of a person or animal and only send alarm signals when it is a boat or person. This way we could minimize unnecessary signals being sent. The buoys will have batteries, also class A to have them last longer but probably also solar power so that we can use energy harvesting as well as batteries in the summer. Class A devices use a protocol similar to ALOHA where there are two receiver windows, first one is 1 second, then the device go back to sleep, then it wakes up a second time for 2 seconds. If an ACK is not received within these windows, the data is resent.

My solution might not be the least expensive one, but I still think it keeps the prices relatively low since LoRa is a rather budget friendly alternative whereas there are no subscriptions fees like there would be if we used cellular network (NB-IoT) or Argos. The main cost is the gateway, the sensors and the buoys. Hopefully overall, the foundation is happy with my solution.